KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2010
Department of Computer Science and Engineering
CSE 4215
Information Security and Control

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What is public key encryption? How does it differ from private key encryption? (10)
   b) Define homomorphic encryption. RSA and ElGamal cryptosystems are homomorphic – explain. (12)
   c) Why is RSA encryption algorithm deterministic? In RSA, how $ed=1$? [Where e and d have their usual meanings]. (13)

2. a) ElGamal encryption algorithm is probabilistic – explain. (11)
   b) Explain the steps of AES encryption algorithm. (12)
   c) Define hash function. What are the applications of hash function in cryptography? (07)
   d) Explain triple DES mechanism. (05)

3. a) What are the requirements for the security of database? Explain them precisely. (10)
   b) Define sensitive data. What are the factors that can make data sensitive? Discuss them. (12)
   c) What are the parameters that can make a network vulnerable? Among them, explain 'anonymity' and 'sharing'. (13)

4. a) Explain RSA digital signature algorithm with an example. (12)
   b) What is the limitation of existing internet system? Explain denial-of-service (DoS) attacks. (10)
   c) Define authentication. What are the threats to e-mail? Explain. (13)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Make a comparison between copyrights and patents. (09)
   b) How trade secret protection is vanished by 'reverse engineering'? Explain. (08)
   c) How does operation system offer protection at several levels? (08)
   d) Explain paging combined with segmentation. What are the advantages and disadvantages of this technique for protecting in general purpose operating system? (10)

6. a) "Information, Security, and Control are three main ingredients of a secure system" – justify the statement. (09)
   b) What are the complementary goals of information security? Describe them vividly. (10)
   c) What is a threat in a computing system? Describe different types of threats with proper example(s). (10)
   d) Differentiate between AES and DES. (06)

7. a) What is data security? How does it differ from network security? (08)
   b) What is buffer overflow problem? How it affects in security implication of a system? (08)
   c) Differentiate between virus and worm. (06)
   d) What is virus gain control? Explain with proper figure(s). (08)
   e) How a virus can be identified? (05)

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8. a) What is directory system? Describe its advantages and disadvantages with appropriate figure(s). (12)
   b) How does military security model fulfill need–to–know rule and security requirements? (14)
   c) What is lattice? How is military security model represented by lattice? (09)

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